Dataset and benchmark limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Dataset and benchmark limitations before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sensitivity analysis

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Dataset and benchmark limitations trace and record the first step you can explain without help.

Use W2: Trace the worked case for Dataset and **benchmark** limitations. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Dataset and benchmark limitations explanation against the supplied facts before discussing it.

Use the worked case for Dataset and **benchmark** limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Dataset and benchmark limitations facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Dataset and benchmark limitations facts and keep your reasoning.

Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Dataset and benchmark limitations facts and keep your reasoning.

A report omits this qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Dataset and benchmark limitations facts and keep your reasoning.

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Dataset and benchmark limitations facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Dataset and benchmark limitations to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled dataset and **benchmark** limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of dataset and **benchmark** limitations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Dataset and benchmark limitations

> A benchmark can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos. Explain the labelled step before checking the changed condition.

A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

Indoor **benchmark**: 9/10 correct. No outdoor cases are present. The same model is intended for both.

AI ACADEMY / AI-7/18

Benchmark conditions

	Correct	Total
Indoor benchmark	9	10
Added outdoor slice	2	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/18

Dataset and benchmark limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name measured setting

All ten benchmark images are clean indoor photos.

2. Report the result

The observed indoor score is 90%.

3. Name missing evidence

Outdoor shade performance remains unmeasured.

CHANGE ONE THING

Add four reserved outdoor cases, of which two are correct.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Report the indoor **benchmark** with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Outdoor shade performance remains unmeasured.

1. Report the result

2. Name measured setting

3. Name missing **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Name missing **evidence** Outdoor shade performance remains unmeasured. A **benchmark** represents particular tasks and collection conditions.

High performance on it cannot automatically establish fitness for a different deployment setting.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add four reserved outdoor cases, of which two are correct. Which conclusion is supported by the supplied facts?

1. Report the outdoor slice as 2/4=50%, separately from indoor results.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report the outdoor slice as 2/4=50%, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

Repair a missing building block

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the **mechanism** and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject.

Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

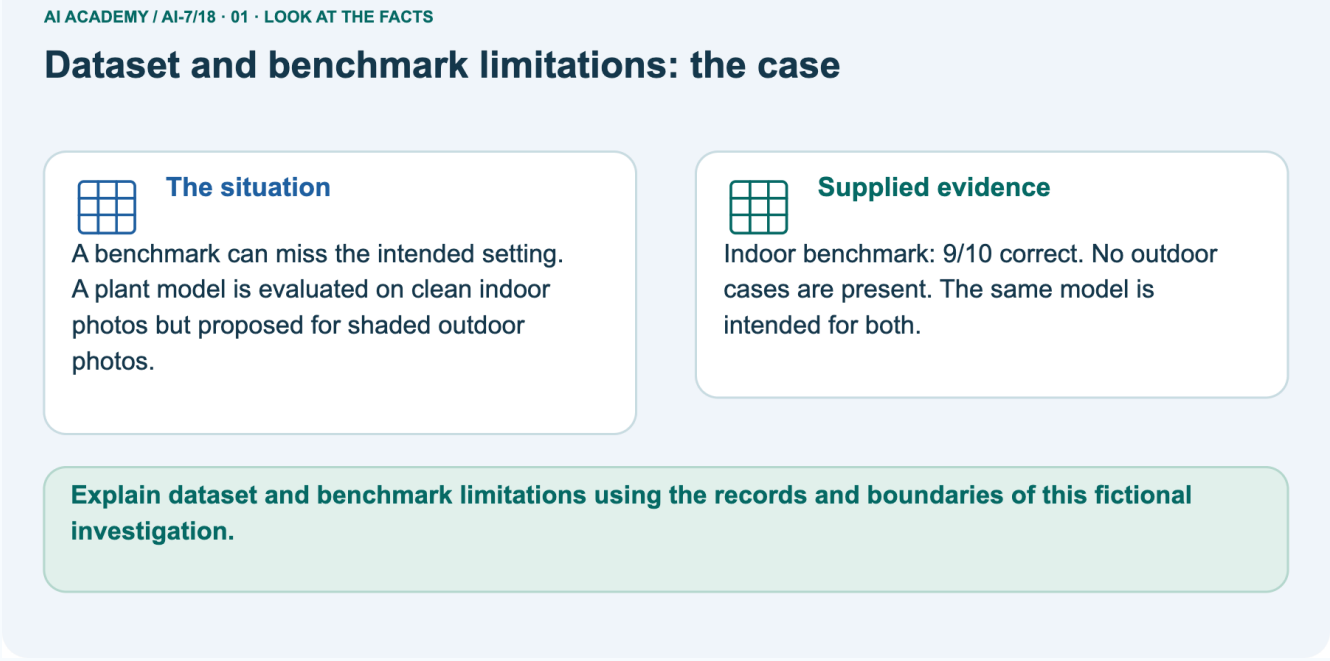
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Dataset and benchmark limitations picture from its labels, then record a relevant limit.

01. Dataset and benchmark limitations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

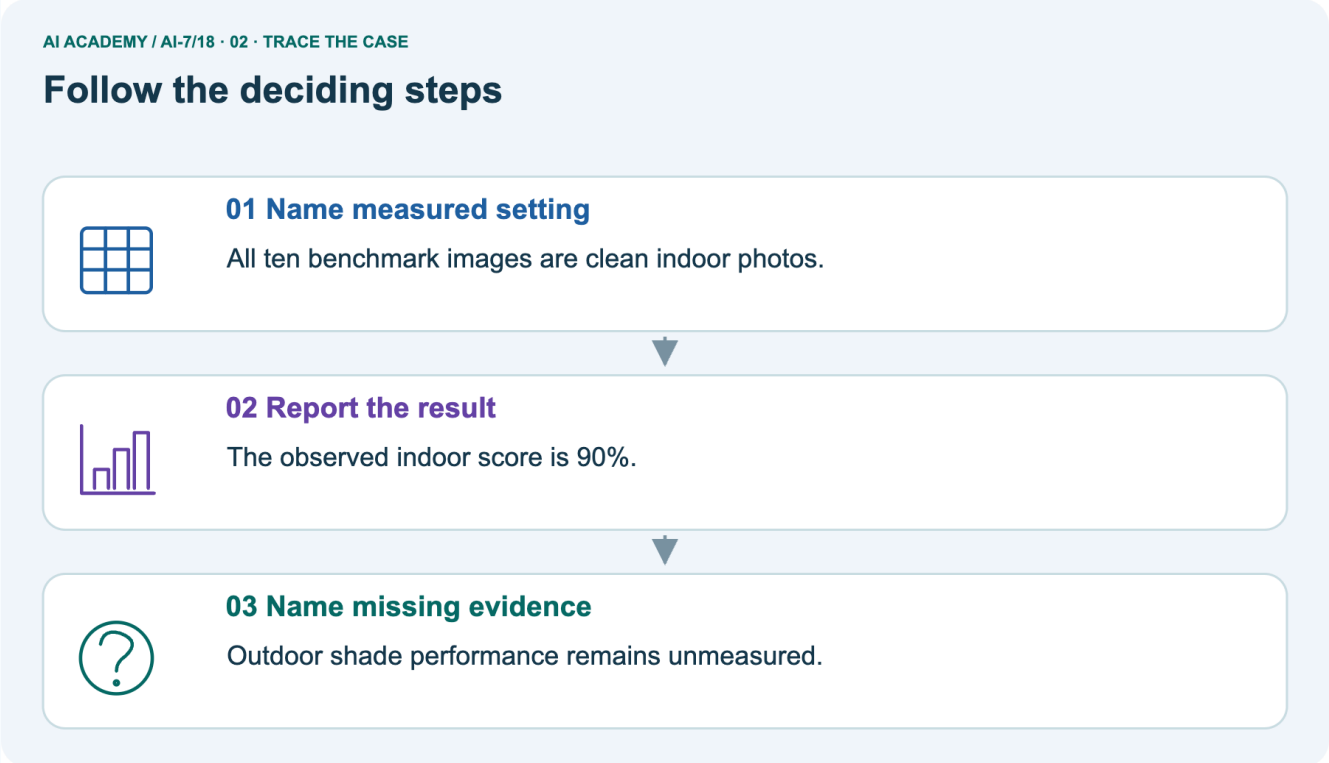
A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos. Supplied facts: Indoor **benchmark**: 9/10 correct. No outdoor cases are present.

The same model is intended for both.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



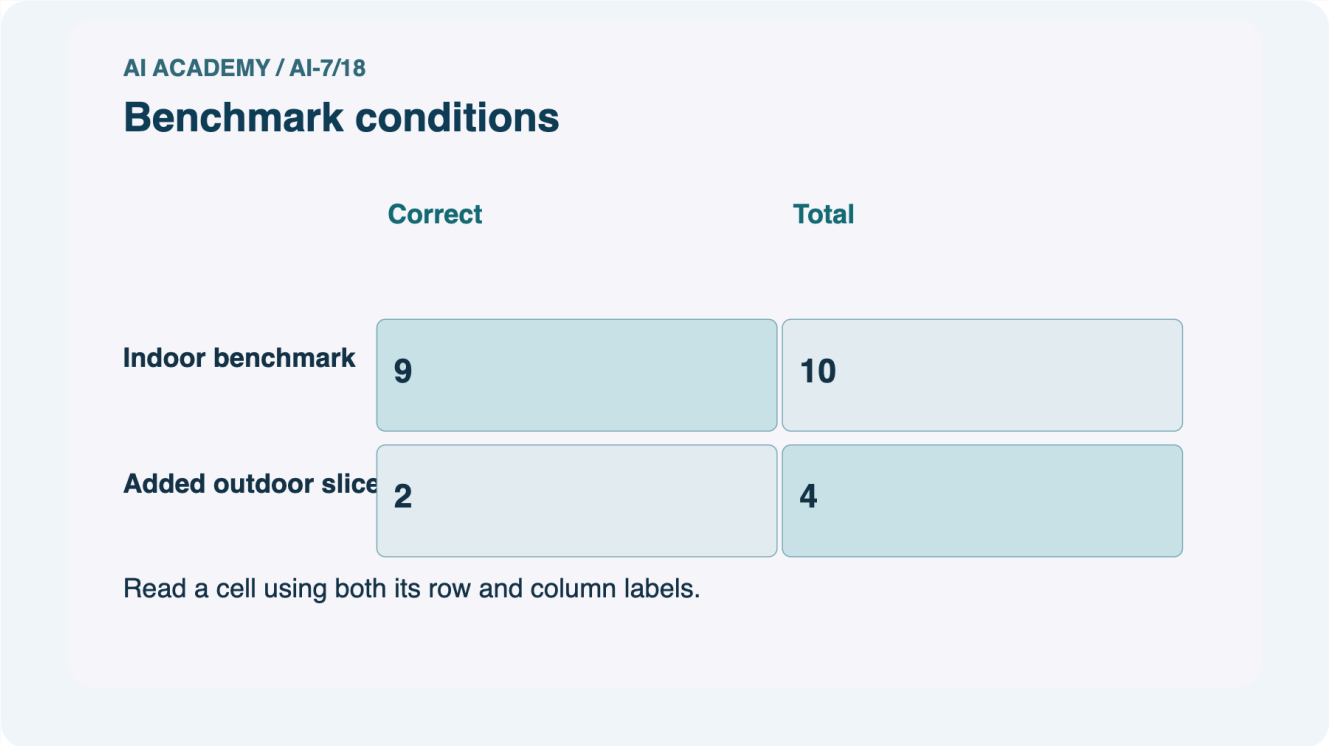
Follow the labelled picture with the matching explanation. Open the vector diagram

Name measured setting: All ten **benchmark** images are clean indoor photos. → Report the result: The observed indoor score is 90%. → Name missing **evidence**: Outdoor shade performance remains unmeasured.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Benchmark conditions



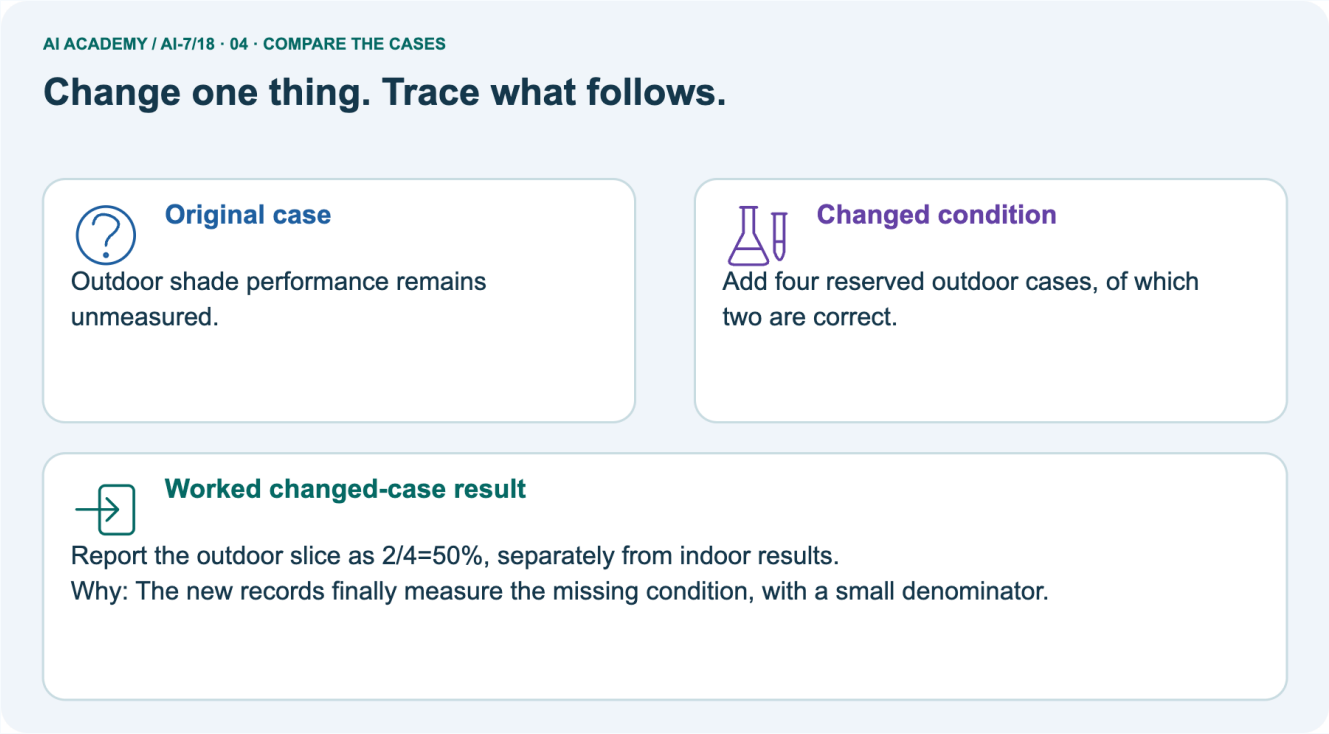
Follow the labelled picture with the matching explanation. Open the vector diagram

Benchmark conditions. Indoor **benchmark**: Correct = 9, Total = 10. Added outdoor slice: Correct = 2, Total = 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

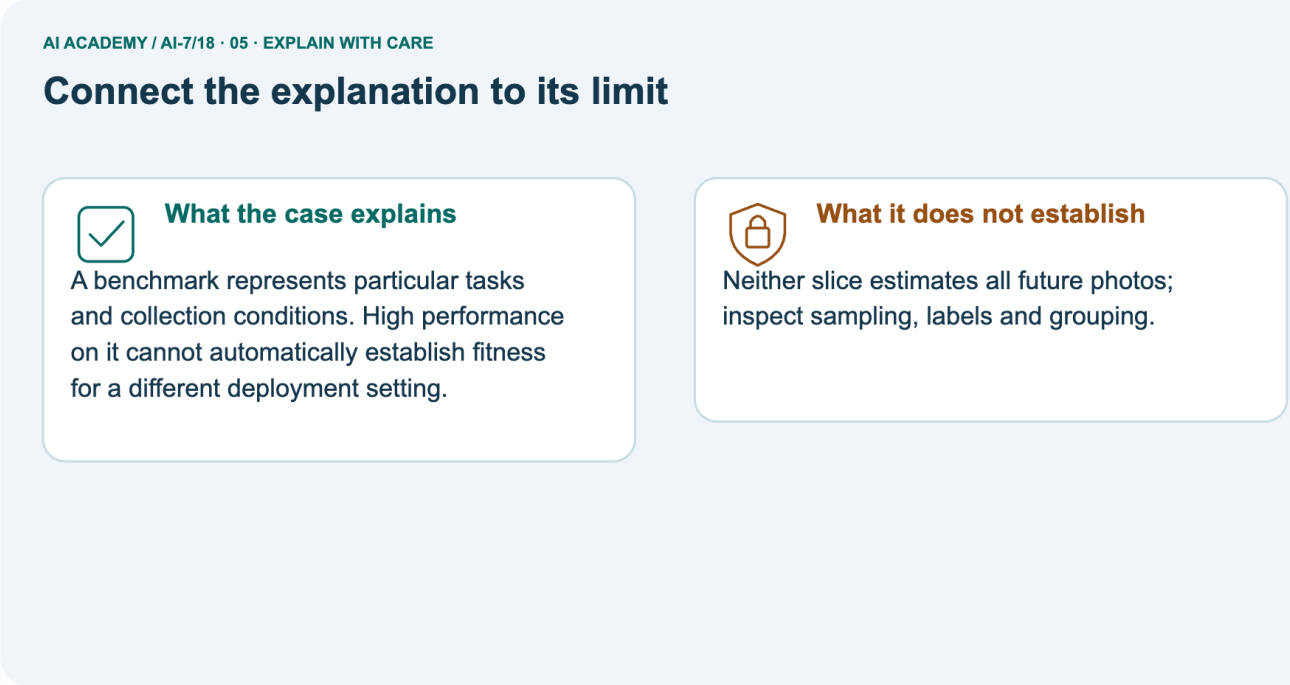
Original case: Outdoor shade performance remains unmeasured. Changed condition: Add four reserved outdoor cases, of which two are correct. Expected result: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

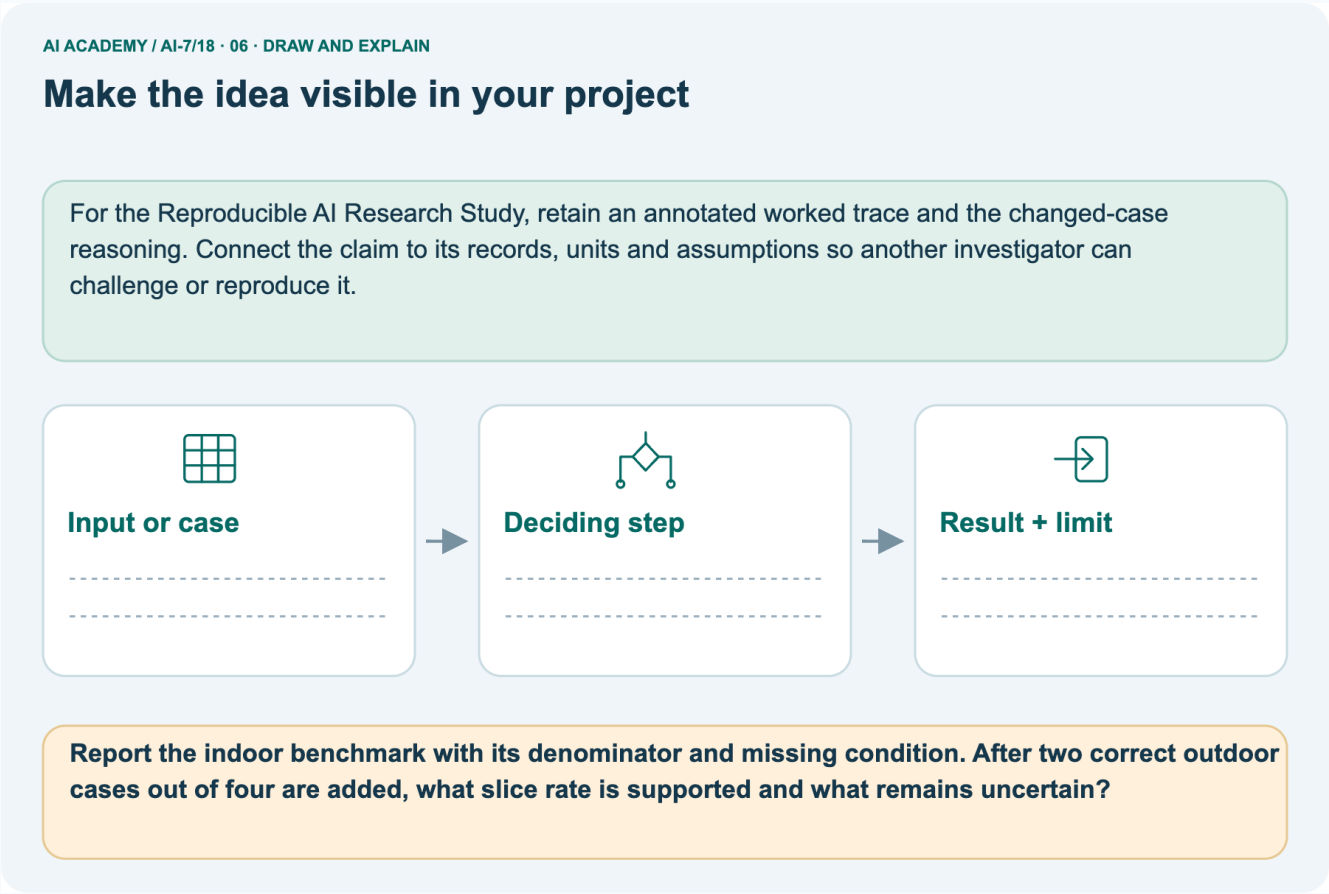
A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Limit: Neither slice estimates all future photos; inspect sampling, labels and grouping.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Report the indoor **benchmark** with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.